

WAREHOUSE ROBOT NAVIGATION SYSTEM

STORY BACKGROUND: A large e-commerce warehouse is adopting autonomous robots to improve efficiency in handling customer orders. These robots are responsible for locating items, picking them from shelves, and delivering them to packing stations. The warehouse is divided into multiple zones, with thousands of items stored across different shelves. Every day, a high number of orders are received, requiring fast and accurate coordination between robots. To ensure smooth operations, the system must:

1. Manage incoming orders efficiently
2. Assign tasks fairly among robots
3. Help robots navigate through the warehouse
4. Quickly locate items
5. Allow robots to return safely after completing tasks

To achieve this, the system is designed using several fundamental data organization techniques. Incoming orders are handled in a structured sequence to ensure fair processing. Robot assignments follow a rotating mechanism so that workload is evenly distributed. The warehouse layout is modeled in a hierarchical form to represent zones, aisles, and shelves, enabling efficient navigation and path planning. Additionally, robot movements are recorded step-by-step, allowing the system to retrace and reverse the path taken. This ensures that once a task is completed, the robot can return to its starting point safely and efficiently. One special feature of the system is the ability for robots to trace back their movement path and return using the exact reverse route, ensuring efficiency and avoiding navigation errors. Overall, the system is developed by applying key concepts such as stack, queue, circular queue, and tree structures to ensure efficiency, fairness, and accuracy in warehouse operations.

FULL SYSTEM WORKFLOW: All modules must work together as one complete system. The full system flow as below:

1. A new order is received
2. The order is stored and prepared for processing
3. A robot is assigned to the task
4. The system identifies the item location
5. A route is generated for the robot
6. The robot moves step-by-step to the item
7. After completing the task, the robot returns using the reverse path

System Tasks That Can Be Allocated to Each Member for This Assignment

Managing a warehouse robot navigation system requires coordinating multiple complex operations to ensure smooth and efficient order fulfilment. Each component of the system involves specific tasks that can be optimized using suitable data structures such as Stack, Queue, Circular Queue, and Tree structures. The system is not limited to the following modules; however, the tasks outlined below represent the core functionalities essential for the successful operation of the warehouse system:

TASK 1: ORDER MANAGEMENT MODULE

The Order Management Module is responsible for handling all incoming customer orders within the warehouse. It ensures that every order is recorded, organized, and processed in a structured and fair manner based on its arrival time.

Functional Requirements:

- Accept and record new customer orders into the system
- Maintain an ordered list of all incoming requests
- Process orders sequentially according to their arrival
- Remove orders from the list once they are assigned to robots
- Display current pending and completed orders
- Handle exceptional cases such as empty order lists or system overload

Key Features:

- Ensures fair processing of orders
- Maintains real-time order status updates
- Supports continuous inflow of new orders

Expected Core Output:

- List of pending orders
- Current order being processed
- Completed order

TASK 2: ROBOT ASSIGNMENT MODULE

The Robot Assignment Module manages how tasks are distributed among available robots. It ensures that all robots are utilized efficiently and fairly by assigning tasks in a rotating and balanced manner.

Functional Requirements:

- Maintain a list of all robots and their current status (available/busy)
- Assign tasks to robots in a continuous rotation
- Skip robots that are currently unavailable or under maintenance
- Track task assignments for each robot
- Ensure uninterrupted task allocation without restarting the assignment cycle

Key Features:

- Balanced workload distribution among robots
- Continuous and automated assignment process
- Real-time robot availability tracking

Expected Core Output:

- Robot assignment list
- Current robot handling each task
- Robot status overview

TASK 3: ROBOT NAVIGATION AND PATH TRACKING MODULE

This module controls the movement of robots within the warehouse and ensures accurate navigation. It also enables robots to return safely by retracing their movement path in reverse order.

Functional Requirements

- Record each movement step taken by the robot (e.g., forward, left, right)
- Store the full path from starting point to destination
- Allow the robot to reverse its path step-by-step after completing a task
- Handle navigation issues such as obstacles or incorrect paths
- Simulate robot movement through logs or step-by-step visualization

Key Features:

- Accurate path tracking and movement recording
- Reverse navigation capability for return journeys
- Basic obstacle handling through backtracking

Expected Core Output:

- Forward movement path
- Reverse path for returning
- Complete navigation log

TASK 4: ITEM SEARCH AND MANAGEMENT MODULE (OPTIONAL)

The Item Search and Management Module is responsible for storing and managing all item-related information in the warehouse. It allows quick and efficient retrieval of item locations when required.

Functional Requirements:

- Store item details such as Item ID, name, and location
- Insert new items into the system
- Search for items based on specific criteria (e.g., ID or name)
- Update or delete item records when necessary
- Display items in a structured and sorted format

Key Features:

- Fast and efficient item lookup
- Organized storage of item information
- Easy management of item records

Expected Core Output:

- Search results for requested items
- Updated item database
- Organized item listing

TASK 5: WAREHOUSE LAYOUT AND NAVIGATION MODULE (OPTIONAL)

This module represents the physical structure of the warehouse and supports navigation by defining relationships between zones, aisles, and shelves.

Functional Requirements:

- Model the warehouse layout into structured sections (zones, aisles, shelves)
- Define connections between different locations
- Provide navigation routes from one point to another
- Support traversal through all warehouse sections
- Integrate with the robot navigation module for path planning

Key Features:

- Structured representation of warehouse layout
- Efficient route generation
- Scalable design for large warehouse environments

Expected Core Output:

- Visual or textual representation of warehouse layout
- Path between selected locations
- Traversal results